

Name

ID#

Math 150 Spring 2025

Final Exam

1) Evaluate the limit two ways

a) Describe exactly what the limit represents in calculus terms.

b) Use L'Hôpital's rule if possible (explain why or why you are not able to use the rule) ✓

The limit represents
the derivative of a
function $f(x)$ ✓

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos(x)} - 1}{x - \frac{\pi}{2}}$$

$$\frac{f(a+h) - f(a)}{h}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos(\frac{\pi}{2})} - 1}{\frac{\pi}{2} - \frac{\pi}{2}} = \frac{0}{0} \checkmark$$

L'H

$$\frac{-\sin(x) e^{\cos x}}{1}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos(x)} - 1}{x - \frac{\pi}{2}} = \boxed{-1} \checkmark$$

$$\begin{aligned} \frac{-e^{\cos(x)}}{-\sin(x)} &= \frac{e^{\cos(\frac{\pi}{2})}}{-\sin(\frac{\pi}{2})} = \frac{1}{-1} = -1 \\ &= -\sin(\frac{\pi}{2}) e^{\cos(\frac{\pi}{2})} \\ &= -(1) e^0 \\ &= -1 \cdot 1 \\ &= -1 \end{aligned}$$

① 5/5

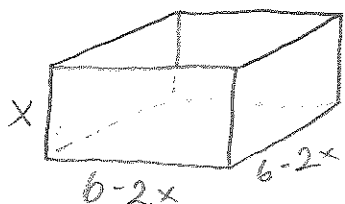
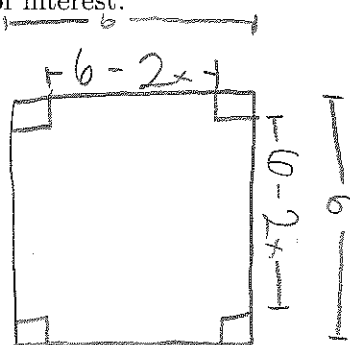
② 5/5

③ 5/5

⑤ 5/5

⑥ 5/5

2) A sheet of paper measuring 6in by 6in is constructed into an open top box by removing four corners. Find the exact dimensions of the box with maximum volume. State the objective function and interval of interest.



$$V = l \cdot w \cdot h$$

$$V = (6-2x)(6-2x)x \quad [0, 3]$$

$$= (36 - 12x - 12x + 4x^2)x$$

$$= (36 - 24x + 4x^2)x$$

$$V = 36x - 24x^2 + 4x^3$$

$$V' = 36 - 48x + 12x^2$$

$$0 = 12(3 - 4x + x^2)$$

$$0 = x^2 - 4x + 3$$

$$(x-3)(x-1)$$

$$x=3, x=1$$

$$l = 4 \text{ in}$$

$$w = 4 \text{ in}$$

$$h = 3 \text{ in}$$

Maximized Volume
is 48 in^3

$$V = (6-2x)^2 x \quad \cdot \text{use } 1 \text{ for } l/w, 3 \text{ for } h$$

$$(6-2(1))^2 \cdot 3$$

$$(6-2)^2 \cdot 3$$

$$4^2 \cdot 3$$

$$16 \cdot 3$$

$$48 \text{ in}^3$$

$$V = (6-2x)^2 x \quad \cdot \text{use } 3 \text{ for } l/w, 1 \text{ for } h$$

$$(6-2(3))^2 \cdot 1$$

$$(6-6)^2 \cdot 1$$

$$V = 0$$

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3) A model rocket is fired vertically upward with an initial velocity of 49 m/s from a tower 150 meters high.

✓ a) How long will it take for the rocket to reach its maximum height?

b) What is the maximum height?

c) How long will it take for the rocket to hit the ground?

d) What will be its speed at impact?

$$D = RT$$

5 seconds

$$g = 9.8 \text{ m/s}^2$$

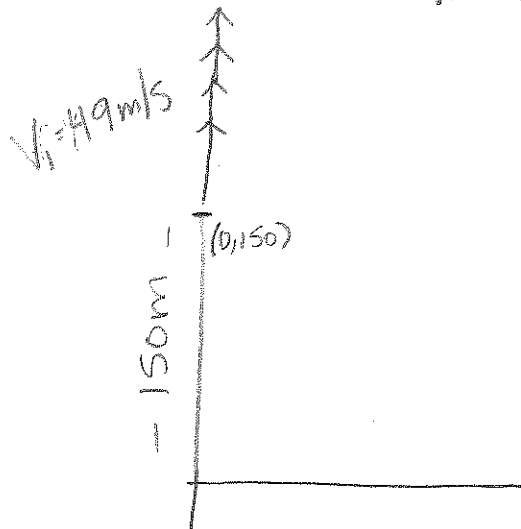
272.5 m

12.45 sec

-73.61 m/s

$$\text{Position} = s(t)$$

$$\text{Velocity} = v(t)$$



$$\int a(t) = -9.8 \text{ m/s}^2$$

$$v(t) = -9.8t + C$$

$$v(0) = -9.8(0) + C$$

$$49 = 0 + C$$

$$v(t) = -9.8t + 49$$

$$s(t) = -4.9t^2 + 49t + 150$$

$$s(0) = 150 \text{ m}$$

$$v(0) = 49 \text{ m/s}$$

$$\int v(t) = -9.8t + 49$$

$$s(t) = \frac{-9.8t^2}{2} + 49t + C$$

$$s(0) = -4.9t^2 + 49t + C$$

$$150 = 0 + 0 + C$$

$$150 = C$$

$$v(t) = -9.8t + 49$$

$$0 = -9.8t + 49$$

$$9.8t = 49$$

$$t = \frac{49}{9.8} = 5 \text{ seconds}$$

↑
max

$$s(5) = -4.9(5)^2 + 49(5) + 150$$

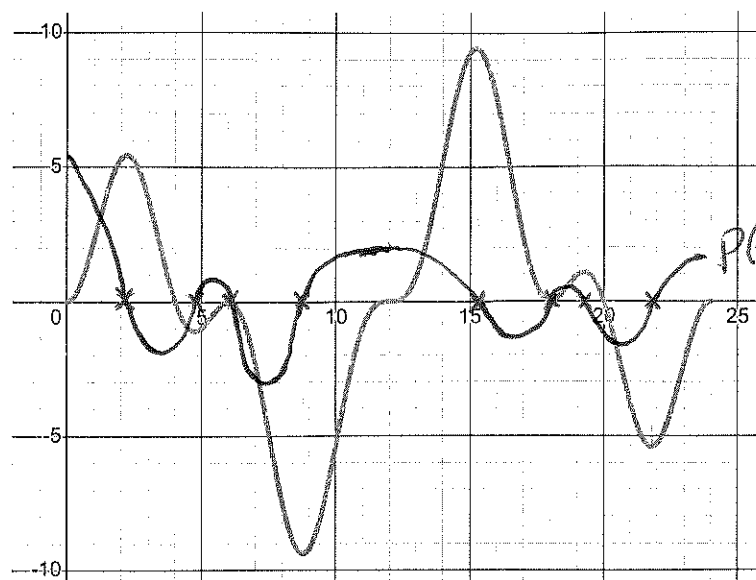
C & D on Black





4) Given the graph of $E(t)$ over a 24 hour period below, and that power $P(t) = E'(t)$.

Graph power $P(t)$ and estimate the time(s) is the power is maximized and minimized.



Graph the
derivative!

$P(t) = E'(t)$

Power maximized at $t \approx 5.5, 12.5, 18.5$
minimized at $t \approx 3.5, 7.5, 16.5, 20.5$



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5) A block of ice in the shape of a perfect cube is melting. Each side of the ice cube is decreasing at a constant rate of 5 mm per minute.

- a) How quickly is the surface area of the ice cube shrinking when each side is 50 mm? $SA = 6s^2$
- The surface area is shrinking at a rate of $3000 \frac{\text{mm}^2}{\text{min}}$
- b) How quickly is the volume of the ice cube shrinking when each side is 50 mm? $V = s^3$
- The volume is shrinking at a rate of $37,500 \frac{\text{mm}^3}{\text{min}}$

$$\frac{ds}{dt} = -5 \frac{\text{mm}}{\text{min}}$$

$$SA = 6s^2$$

$$\begin{aligned} \frac{dSA}{dt} &= 12s \frac{ds}{dt} = 12(50 \text{ mm}) \left(-5 \frac{\text{mm}}{\text{min}} \right) \\ &= -3000 \frac{\text{mm}^2}{\text{min}} \end{aligned}$$

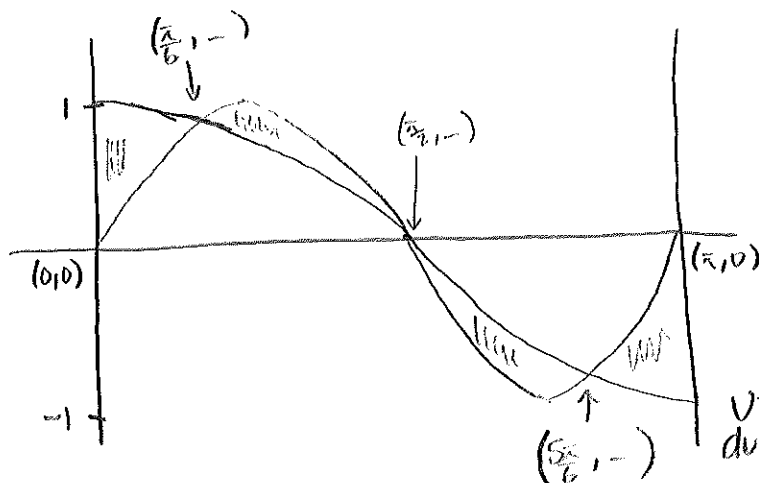
$$V = s^3$$

$$\begin{aligned} \frac{dV}{dt} &= 3s^2 \frac{ds}{dt} = 3(50 \text{ mm})^2 \left(-5 \frac{\text{mm}}{\text{min}} \right) \\ &= -37500 \frac{\text{mm}^3}{\text{min}} \end{aligned}$$

$[0, \pi]$



6) Find the area between the curves $y = \sin(2x)$, $y = \cos(x)$, $x = 0$, and $x = \pi$



$$A = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

$$= 1$$

$$u = \sin x$$

$$du = \cos x \, dx$$

$$x=0, u=\sin(0)=0$$

$$x=\frac{\pi}{6}, u=\sin(\frac{\pi}{6})=\frac{1}{2}$$

$$x=\frac{\pi}{2}, u=\sin(\frac{\pi}{2})=1$$

$$x=\frac{5\pi}{6}, u=\sin(\frac{5\pi}{6})=\frac{1}{2}$$

$$x=\pi, u=\sin(\pi)=0$$

$$\sin(2x) = \cos(x)$$

$$2\sin(x)\cos(x) = \cos(x)$$

$$\frac{2\sin(x)}{2} = \frac{1}{2}$$

$$\sin(x) = \frac{1}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\cos(x) = 0$$

$$x = \frac{\pi}{2}$$

$$2\sin x \cos x = 0$$

$$\sin x \cos x = 0$$

$$x = \frac{\pi}{2}$$

$$A = \int_0^{\frac{\pi}{6}} (\cos(x) - \sin(2x)) \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\sin(2x) - \cos(x)) \, dx + \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} (\cos(x) - \sin(2x)) \, dx + \int_{\frac{5\pi}{6}}^{\pi} (\sin(2x) - \cos(x)) \, dx$$

$$A = \int_0^{\frac{\pi}{6}} (\cos x - 2\sin x \cos x) \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2\sin x \cos x - \cos x) \, dx + \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} (\cos x - 2\sin x \cos x) \, dx + \int_{\frac{5\pi}{6}}^{\pi} (2\sin x \cos x - \cos x) \, dx$$

$$A = \int_0^{\frac{\pi}{6}} (1 - 2\sin x) \cos x \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2\sin x - 1) \cos x \, dx + \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} (1 - 2\sin x) \cos x \, dx + \int_{\frac{5\pi}{6}}^{\pi} (2\sin x - 1) \cos x \, dx$$

$$A = \int_0^{\frac{\pi}{6}} (1 - 2\sin x) \cos x \, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2\sin x - 1) \cos x \, dx + \int_{\frac{\pi}{2}}^{\frac{5\pi}{6}} (1 - 2\sin x) \cos x \, dx + \int_{\frac{5\pi}{6}}^{\pi} (2\sin x - 1) \cos x \, dx$$

$$A = \int_0^{\frac{1}{2}} (1 - 2u) \, du + \int_{\frac{1}{2}}^1 (2u - 1) \, du + \int_1^{\frac{1}{2}} (1 - 2u) \, du + \int_{\frac{1}{2}}^0 (2u - 1) \, du$$

$$\left(u - \frac{2u^2}{2} \Big|_0^{\frac{1}{2}} \right) + \left(\frac{2u^2}{2} - u \Big|_{\frac{1}{2}}^1 \right) + \left(u - \frac{2u^2}{2} \Big|_1^{\frac{1}{2}} \right) + \left(\frac{2u^2}{2} - u \Big|_{\frac{1}{2}}^0 \right)$$

$$\left(\frac{1}{2} - \frac{1}{2} \Big|_0^{\frac{1}{2}} \right) + \left(u^2 - u \Big|_{\frac{1}{2}}^1 \right) + \left(u - u^2 \Big|_1^{\frac{1}{2}} \right) + \left(u^2 - u \Big|_{\frac{1}{2}}^0 \right)$$

$$\left(\frac{1}{2} - \frac{1}{2} - (0 - 0) \right) + \left(1^2 - 1 - \left(\left(\frac{1}{2} \right)^2 - \frac{1}{2} \right) \right) + \left(\frac{1}{2} - \left(\frac{1}{2} \right)^2 - (1 - 1) \right) + \left(0^2 - 0 - \left(\left(\frac{1}{2} \right)^2 - \frac{1}{2} \right) \right)$$

$$\frac{1}{2} - \frac{1}{4} + \left(-\frac{1}{4} + \frac{1}{2} \right) + \frac{1}{2} - \frac{1}{4} - \left(\frac{1}{4} - \frac{1}{2} \right)$$

$$\frac{2}{4} - \frac{1}{4} - \frac{1}{4} + \frac{2}{4} + \frac{2}{4} - \frac{1}{4} - \frac{1}{4} + \frac{2}{4}$$

$$\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$$

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7) Calculate the surface area of the surface obtained by revolving the curve $y = \frac{x^3}{3}$ around the x -axis from $1 \leq x \leq 2$.

$$S = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx$$

$$= 2\pi \int_1^2 \frac{x^3}{3} \sqrt{1 + (x^2)^2} dx$$

$$= \frac{2\pi}{3} \int_1^2 x^3 \sqrt{1 + x^4} dx$$

$$= \frac{2\pi}{3} \int_1^2 \sqrt{1 + x^4} \cdot x^3 dx$$

$$= \frac{2\pi}{3} \cdot \frac{1}{4} \int_2^{17} \sqrt{u} du$$

$$\frac{\pi}{6} \int_2^{17} u^{\frac{1}{2}} du$$

$$\frac{\pi}{6} \left(\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \bigg|_2^{17} \right)$$

$$\frac{\pi}{6} \left(\frac{2}{3} u^{\frac{3}{2}} \bigg|_2^{17} \right)$$

$$\frac{2\pi}{9} \left(u^{\frac{3}{2}} \bigg|_2^{17} \right)$$

$$\frac{\pi}{9} \left(17^{\frac{3}{2}} - 2^{\frac{3}{2}} \right)$$

$$f(x) = \frac{x^3}{3}$$

$$f'(x) = \frac{1}{3} (x^3)'$$

$$= \frac{1}{3} (3x^2)$$

$$f'(x) = x^2$$

$$u = 1 + x^4$$

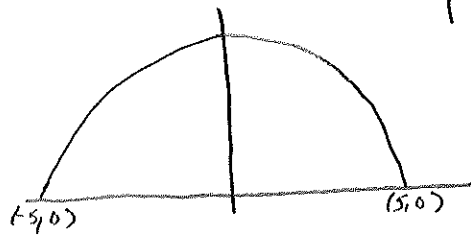
$$du = 4x^3 dx$$

$$\frac{1}{4} du = x^3 dx$$

$$x=2, u = 1 + 2^4 = 1 + 16 = 17$$

$$x=1, u = 1 + 1^4 = 1 + 1 = 2$$

$$A = \frac{\pi}{9} (17\sqrt{17} - 2\sqrt{2}) \text{ units}^2$$



$$r=5$$

$$A_{\frac{1}{2}D} = \pi r^2 \left(\frac{1}{2}\right) = \pi (5)^2 \frac{1}{2} = \left[\frac{25\pi}{2}\right] \checkmark$$

GEOMETRY

$$\approx 39.2699$$

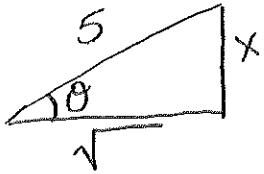
8) Find the exact value of the integral $\int_{-5}^5 \sqrt{25-x^2} dx$ using integration techniques and geometry. $\checkmark$

MID

Use Riemann Sums to approximate the integral with 10 rectangles and use Sigma notation (draw the rectangles on the graph).

(ON THE BACK)

How many rectangles are necessary to approximate to within 0.001 of the exact value?



$$\sqrt{25-x^2} = 5^2 - x^2$$

$$\begin{aligned} \sin \theta &= \frac{x}{5} & \csc \theta &= \frac{5}{x} \\ \cos \theta &= \frac{\sqrt{25-x^2}}{5} & \sec \theta &= \frac{5}{\sqrt{25-x^2}} \\ \tan \theta &= \frac{x}{\sqrt{25-x^2}} & \cot \theta &= \frac{\sqrt{25-x^2}}{x} \end{aligned}$$

$$5 \cos \theta = \sqrt{25-x^2}$$

$$5 \sin \theta = x$$

$$5 \cos \theta d\theta = dx$$

$$\sin \theta = \frac{x}{5}$$

$$\theta = \arcsin\left(\frac{x}{5}\right)$$

$$= \arcsin\left(\frac{5}{5}\right) = \arcsin(1)$$

$$x = \frac{5}{2}$$

$$= \arcsin\left(-\frac{5}{5}\right) = \arcsin(-1)$$

$$x = -\frac{5}{2}$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 5 \cos \theta \cdot 5 \cos \theta d\theta$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 25 \cos^2 \theta d\theta$$

$$25 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$25 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos(2\theta)) d\theta$$

$$\frac{25}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \cos(2\theta)) d\theta$$

$$\frac{25}{2} \left(\theta + \frac{\sin(2\theta)}{2} \right) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$\frac{25}{2} \left(\frac{\pi}{2} + \frac{\sin(2 \cdot \frac{\pi}{2})}{2} - \left(-\frac{\pi}{2} + \frac{\sin(2 \cdot -\frac{\pi}{2})}{2} \right) \right)$$

$$\frac{25}{2} \left(\frac{\pi}{2} + \frac{\sin(\pi)}{2} - \left(-\frac{\pi}{2} + \frac{\sin(-\pi)}{2} \right) \right)$$

$$\frac{25}{2} \left(\frac{\pi}{2} + \frac{0}{2} - \left(-\frac{\pi}{2} + \frac{0}{2} \right) \right)$$

$$\frac{25}{2} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{25}{2} \left(\frac{2\pi}{2} \right) = \left[\frac{25\pi}{2} \right] \checkmark$$

$$f(y) = \frac{\sin(2\theta)}{2}$$

$$f'(y) = \frac{\cos(2\theta) \cdot 2}{2}$$

Goal

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9) Let $y = \tan^{-1}(x)$. Implicitly differentiate to derive the derivative $\frac{d}{dx}(\tan^{-1}(x))$.

$$(\tan(y) = x)' \quad x = \tan y, \quad \sec^2 y = 1 + \tan^2 y$$

$$\sec^2(y) \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \cos^2(y) = 1 - \sin^2 y$$

$$\frac{d}{dx}(\arctan(x)) = \frac{1}{1+x^2}$$

$$\frac{dy}{dx} = \frac{1}{\sec^2(y)}$$

$$= \frac{1}{1+\tan^2 y}$$

$$\boxed{\frac{1}{1+x^2}}$$

✓

9b) Let $y = \cosh^{-1}(x)$. Implicitly differentiate to derive the derivative $\frac{d}{dx}(\cosh^{-1}(x))$.

$$(\cosh(y) = x)'$$

$$\sinh(y) \frac{dy}{dx} = 1$$

$$\cosh^2 y - \sinh^2 y = 1$$

$$\cosh^2 y - 1 = \sinh^2 y$$

$$1 \sqrt{\cosh^2 y - 1} = \sinh y$$

$$\sinh y = \sqrt{\cosh^2 y - 1}$$

Goal

$$\Rightarrow \frac{1}{\sqrt{x^2 - 1}}$$

$$\frac{dy}{dx} = \frac{1}{\sinh(y)}$$

$$= \frac{1}{\sqrt{\cosh^2 y - 1}} = \boxed{\frac{1}{\sqrt{x^2 - 1}}}$$

✓

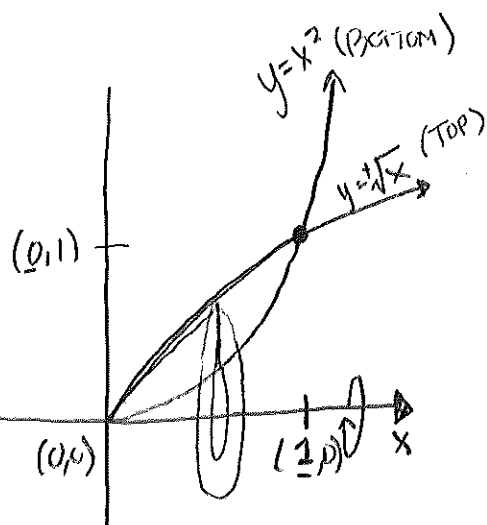
$$= \frac{1}{\frac{e^y - e^{-y}}{2}} = \frac{2}{e^y - e^{-y}}$$

$$= \frac{2}{e^y - e^{-y}} \cdot \frac{e^y}{e^y} = \frac{2e^y}{e^{2y} - 1}$$

$$= \boxed{\frac{2e^y}{e^{2y} - 1}}$$

$$y = x^2, y = \sqrt{x}$$

10) Find the volume generated by revolving the region bounded by the curves $y = x^2$ and $x = y^2$ about the x -axis first using washers, then use shells. Show the two integrals are equal.



WASHERS $= \pi \int_a^b ((f(x))^2 - (g(x))^2) dx$

$$V = \pi \int_0^1 (\sqrt{x})^2 - (x^2)^2 dx$$

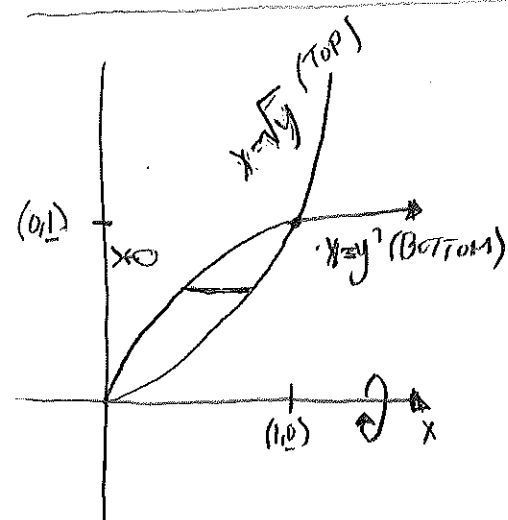
$$= \pi \int_0^1 x - x^4 dx$$

$$= \pi \left(\frac{x^2}{2} - \frac{x^5}{5} \right) \Big|_0^1$$

$$= \pi \left(\frac{1^2}{2} - \frac{1^5}{5} - \left(\frac{0^2}{2} - \frac{0^5}{5} \right) \right)$$

$$= \pi \left(\frac{1}{2} - \frac{1}{5} \right)$$

$$= \pi \left(\frac{5}{10} - \frac{2}{10} \right) = \boxed{\frac{3\pi}{10}} \checkmark$$



Shells $= 2\pi \int_a^b y(g(y)) dy$

$$V = 2\pi \int_0^1 y(\sqrt{y} - y^2) dy$$

$$= 2\pi \int_0^1 y(y^{\frac{1}{2}} - y^2) dy$$

$$= 2\pi \int_0^1 y^{\frac{3}{2}} - y^3 dy$$

$$= 2\pi \left(\frac{y^{\frac{5}{2}}}{\frac{5}{2}} - \frac{y^4}{4} \right) \Big|_0^1$$

$$= 2\pi \left(\frac{2}{5} y^{\frac{5}{2}} - \frac{1}{4} y^4 \right) \Big|_0^1$$

$$= 2\pi \left(\frac{2}{5} (1)^{\frac{5}{2}} - \frac{1}{4} (1)^4 - \left(\frac{2}{5} (0)^{\frac{5}{2}} - \frac{1}{4} (0)^4 \right) \right)$$

$$= 2\pi \left(\frac{2}{5} - \frac{1}{4} \right)$$

$$= 2\pi \left(\frac{8}{20} - \frac{5}{20} \right)$$

$$= 2\pi \left(\frac{3}{20} \right) = \boxed{\frac{3\pi}{10}} \checkmark$$